



Solution Set

Question 1

For

$$\int \frac{7}{2x+3} dx$$

use

$$u = 2x + 3, \quad du = 2dx$$

$$\int \frac{7}{2x+3} dx = \frac{7}{2} \log|2x+3| + C$$

Question 2

For

$$\int \sqrt{7x+9} dx$$

use

$$u = 7x + 9$$

$$\int (7x+9)^{1/2} dx = \frac{2}{21} (7x+9)^{3/2} + C$$

Question 3

For

$$\int (2x+5)(x^2+5x)^8 dx$$

use

$$u = x^2 + 5x, \quad du = (2x+5)dx$$

$$\int u^8 du = \frac{(x^2+5x)^9}{9} + C$$

Question 4

For

$$\int \frac{\sin(\log x)}{x} dx$$

use

$$u = \log x, \quad du = \frac{dx}{x}$$

$$\int \sin u du = -\cos(\log x) + C$$

Question 5

For

$$\int x \sqrt{4-x} \, dx$$

use

$$\begin{aligned} u &= 4 - x, \quad x = 4 - u, \quad dx = -du \\ \int x \sqrt{4-x} \, dx &= - \int (4-u) u^{1/2} \, du \\ &= -\frac{8}{3}(4-x)^{3/2} + \frac{2}{5}(4-x)^{5/2} + C \end{aligned}$$

Question 6

For

$$\int (\log x)^3 \, dx$$

integrate by parts with

$$\begin{aligned} u &= (\log x)^3, \quad dv = dx \\ \int (\log x)^3 \, dx &= x \{ (\log x)^3 - 3(\log x)^2 + 6\log x - 6 \} + C \end{aligned}$$

Question 7

For

$$\int \left(\frac{\log x}{x} \right)^2 \, dx$$

there is no elementary antiderivative. With

$$u = \log x$$

we get

$$\int u^2 e^{-u} \, du = -(u^2 + 2u + 2)e^{-u} + C$$

so

$$\int \left(\frac{\log x}{x} \right)^2 \, dx = -\frac{(\log x)^2 + 2\log x + 2}{x} + C$$

Question 8

Area under

$$y = \sqrt{x}$$

from $x = 0$ to $x = 4$ is

$$\int_0^4 \sqrt{x} \, dx = \left[\frac{2}{3} x^{3/2} \right]_0^4 = \frac{2}{3}(8) = \frac{16}{3}$$

Question 9

The region bounded by

$$y = x, \quad y = 3x, \quad x = 2$$

is between the two lines for

$$0 \leq x \leq 2$$
$$A = \int_0^2 (3x - x) dx = \int_0^2 2x dx = [x^2]_0^2 = 4$$